HALL KHOJ FINDER: A FULL-STACK 3D EVENT SERVICES & VENUE BOOKING MARKETPLACE

A Major Capstone Project Synopsis Submitted in Partial Fulfillment of Academic Requirements

Academic Curriculum Parameter	Project Metadata Specification
Project Title	Hall Khoj Finder — Premium 3D Event Services Marketplace
Candidate / Student Name	Project Developer (Hall Khoj Development Team)
Degree & Specialization	Bachelor of Technology (B.Tech in Computer Science & Engineering) / BCA / MCA
Department & Faculty	Department of Computer Science & Engineering / Information Technology
Institutional Affiliation	College of Engineering & Technology / State Technological University
Academic Session & Term	2025 – 2026 (Final Year Major Degree Project)
Core Technological Domains	Full-Stack Web Engineering, 3D Computer Graphics (WebGL) & Distributed Cloud APIs
System Architecture Model	Decoupled RESTful MVC Monolith (Node.js & Express 5 serving HTML5/Three.js)
Primary Database Engines	MongoDB Atlas (Mongoose ODM) with Transactional Local JSON Resilient Failover
Deployment Target	Unified Cloud Container (Node.js + Docker / Cloud Run / Render Platform)

1. Project Abstract

Hall Khoj Finder is an enterprise-grade, full-stack event services marketplace engineered s systemic inefficiencies of the multi-billion-dollar Indian wedding and celebration hospitality celebratory gatherings—such as weddings, receptions, anniversaries, corporate galas, and procurement of multiple independent vendor domains. These encompass banquet halls, marriag tent house infrastructure, floral and stage decorators, catering agencies, DJ sound systems, a Historically, this discovery and reservation workflow has suffered from severe geographical frag 15% to 30% parasitic middleman broker commissions, unverified verbal agreements, and exhaus

To eradicate these deep-seated bottlenecks, Hall Khoj Finder establishes a unified, cloud-nati standardized event service categories. Built upon a high-performance stack comprising H glassmorphism, native vanilla JavaScript (ES6+), Three.js (r128) WebGL graphics, and a Node. introduces an interactive in-browser 3D procedural venue walkthrough. This allows prospecti proportions, grand entrance foyers, central aisles, chandelier-lit stages, banquet table layouts, three realistic time-of-day lighting simulations (Daylight, Sunset, and Night Gala) directly browsers without external plugins.

The platform enforces rigorous data and financial integrity through a conflict-protected bookin level locks, eradicating devastating double-booking incidents during peak auspicious muhurat 25% advance token payment pipeline backed by Razorpay and cryptographically authenticate alongside a grounded natural language AI search engine parsing queries in colloquial Hind hallucinations. Accompanied by a Pan-India localization engine covering 14 Indian regional dashboards for venue operators and administrators, Hall Khoj Finder delivers an autho transformation for celebratory event planning.

2. Academic Significance & Capstone Research Scope

This project serves as an empirical academic investigation into modern distributed web applic decoupled monolith (Node.js/Express 5 paired with Vanilla ES6+ and WebGL) can achieve su load times on rural mobile devices, and zero double-booking concurrency anomalies without i single-page application (SPA) client frameworks.

3. Academic Keywords & Domain Taxonomy

Full-Stack Web Engineering, Three.js WebGL 3D Simulation, Node.js & Express 5 RE Mongoose ODM, Razorpay Payment Gateway, HMAC-SHA256 Cryptographic Verificat (RBAC), Conflict-Protected Booking State Machine, Grounded Natural Language AI Localization (i18n), Software Reliability.

PAGE 2 — INTRODUCTION, PROBLEM STATEMENT & MOTIVATION

4. Industry Background & Economic Magnitude

The Indian wedding, banquet, and celebration hospitality industry constitutes an economic behemoth valued in excess of INR 4,50,000 Crore ($54 Billion USD) annually, exhibiting consistent double-digit year-on-year growth. Across metropolitan centers, tier-2 cities, and semi-urban districts, life-stage celebrations represent profound socio-cultural milestones where families commit substantial lifetime savings to host memorable gatherings. Organizing these complex multi-day events necessitates seamless synchronization among numerous micro, small, and medium enterprise (MSME) service providers—from marriage halls and community banquets to tent house logistics, culinary teams, sound engineers, and transportation providers.

Despite pervasive digital transformation across retail commerce, urban mobility, and hotel travel, the venue booking and event vendor industry in India remains overwhelmingly anchored in antiquated, paper-based, and non-digitized operational methodologies. Finding an appropriate venue still entails word-of-mouth recommendations, navigating chaotic urban traffic for preliminary site inspections, deciphering vague pricing estimates, and negotiating with unregistered neighborhood brokers who monopolize local availability information. This chronic lack of transparency places celebrating families in an intensely vulnerable negotiating position, often leading to unplanned budget overruns, acute emotional distress, and financial exploitation.

5. Comprehensive Problem Statement

The inception of Hall Khoj Finder addresses eight critical systemic deficiencies in the conventional marketplace:

- **1. Extreme Market & Catalog Fragmentation:** Customers must separately discover, negotiate with, and coordinate marriage halls, luxury wedding cars, tent suppliers, caterers, photographers, and DJs across disjointed classifieds, generating massive cognitive and logistical friction.
- **2. Remote Spatial Visualization Void:** Traditional 2D digital photo galleries or social media snapshots fail to convey true spatial volume, ceiling clearance, pillar obstruction, stage focal lines, or dining seating density, forcing families to make repetitive, time-consuming on-site visits.
- **3. Predatory Brokerage Networks & Rate Inflation:** Unregulated local brokers exploit customer desperation during peak auspicious calendar dates (muhurats), inflating venue prices by 15% to 30% without adding tangible service value.
- **4. Concurrency Failures & Double-Booking Disasters:** Paper desk diaries and asynchronous telephone confirmations lack atomic concurrency safeguards, frequently causing catastrophic overlapping reservations for the same hall on the same evening.
- **5. Opaque Pricing & Insecure Cash Advance Practices:** Vendors demand substantial untracked cash advances without legally binding receipts, transparent itemized invoices, or formal cancellation and refund frameworks.
- **6. Fraudulent & Unverified Review Ecosystems:** Public review forums allow fake, incentivized, or competitor-driven ratings because reviews are not linked to verified, completed transactional records.
- **7. Severe Linguistic Exclusivity:** Most software tools operate solely in English, alienating rural families and traditional venue caretakers who speak regional languages.
- **8. Absence of Vendor Business Automation:** Micro-vendors lack affordable booking software, inventory management, and digital calendar blocking tools, leading to operational chaos.

6. Systemic Vulnerabilities of Conventional Offline Operations

A profound vulnerability of conventional operations is the total absence of verifiable auditability. When disputes arise regarding advance refunds, cancellation policies, or promised amenities, families possess zero legal recourse beyond verbal recollection. Furthermore, venue operators suffer substantial revenue losses due to unconfirmed verbal date holds that prospective clients abandon without penalty. Hall Khoj Finder rectifies this dual dilemma by establishing immutable, timestamped digital transaction ledgers.

7. Project Motivation & Socio-Technical Imperatives

The fundamental motivation behind Hall Khoj Finder is to leverage modern full-stack computer science—specifically hardware-accelerated 3D WebGL graphics, event-driven REST APIs, cryptographic financial verification, and generative artificial intelligence—to democratize and formalize a high-stakes consumer sector. By granting grassroots MSME vendors enterprise-grade digital visibility and providing consumers with verifiable pricing, remote 3D exploration, and conflict-protected transactions, the project bridges the digital divide and restores trust to the celebration economy.

PAGE 3 — LITERATURE SURVEY & EXISTING VS PROPOSED SYSTEM

8. Literature Survey & Related Works Benchmarking

A rigorous survey of prevailing digital platforms in the venue discovery and event management sector reveals significant structural and technological limitations when mapped against the practical requirements of Indian celebration logistics:

- **A. Generic Classified Directories (e.g., Justdial, Sulekha):** These platforms function strictly as lead-generation mechanisms. They broadcast prospective client phone numbers to multiple aggressive sales agents without providing real-time availability, spatial visualizers, direct booking capabilities, or transparent pricing structures.
- **B. High-End Wedding Aggregators (e.g., WedMeGood, WeddingWire India):** While aesthetically pleasing, these platforms focus disproportionately on ultra-luxury metropolitan banquet halls and designer vendors. They act as editorial discovery catalogues rather than real-time transactional booking engines, lacking date-level conflict locks, 3D spatial simulations, and localized vernacular language support.
- **C. Global Hospitality Portals (e.g., Airbnb, Booking.com):** Although possessing world-class calendar booking engines, their architecture is optimized for overnight room stays rather than multi-faceted event logistics involving guest seating capacities, catering styles, sound restrictions, and allied event equipment.
- **D. Specialized Event ERP Suites (e.g., Eventbrite, Cvent):** Targeted exclusively at corporate conferences and ticketed festivals; cost-prohibitive licensing models render them unviable for local marriage halls, tent houses, and family celebration budgets.

9. Existing System Workflow & Operational Flaws

In the conventional offline paradigm, a family intending to organize an event must manually compile candidate venues from friends, undertake physical travel across traffic-congested localities over several weekends, and engage in protracted verbal negotiations with venue caretakers. Dates are confirmed via handwritten paper receipts after handing over large cash tokens. This workflow is fundamentally fragile: handwritten diaries are vulnerable to physical loss or water damage, pricing is arbitrary based on perceived client affluence, and verbal promises regarding lighting, generator backup, or parking capacity often evaporate on event day with zero accountability.

10. Proposed System Architectural Paradigm

Hall Khoj Finder introduces a modern, single-window web application that replaces chaotic offline interactions with an automated, auditable digital workflow. Prospective clients discover verified vendors across 19 categories, explore procedural 3D venue layout simulations with dynamic time-of-day lighting, check live date availability, reserve bookings through an automated 25% advance token payment pipeline, and receive digitally signed transaction receipts. Venue owners are equipped with an administrative SaaS suite to manage listings, track occupancy, block dates, and analyze revenues.

11. Comprehensive System Comparison Matrix

Operational Parameter	Existing Traditional System	Proposed Hall Khoj Finder Platform
Market Discovery	Scattered local contacts, word-of-mouth, physical travel	Unified multi-category web portal across 19 event service domains
Spatial Inspection	Mandatory on-site physical travel across congested roads	Interactive in-browser Three.js 3D walkthrough with 3 lighting ambiances
Availability Tracking	Opaque paper desk diaries; frequent double-booking	Real-time color-coded calendar with atomic database-level date locks
Pricing Transparency	Arbitrary quotes influenced by middleman broker cartels	Zero-commission verified base pricing with transparent capacity brackets
Advance Payment	Untracked cash transactions with handwritten slips	Server-calculated 25% token payment via Razorpay with HMAC verification
Linguistic Access	Limited to regional spoken dialects; documentation in English	Native Pan-India localization engine covering 14 Indian languages
Review Verification	Unverified public reviews prone to competitor manipulation	Anti-fraud gated review engine restricted strictly to completed bookings
Business Analytics	Non-existent; owners rely on rough mental estimates	Comprehensive owner SaaS portal tracking 90-day revenue and occupancy
Dispute Mitigation	Informal verbal arguments with zero legal trail	Immutable financial ledger, audit trails, and digital invoice generation
Vendor Integration	Disconnected contracts with separate independent parties	Integrated multi-category booking with unified customer dashboard

PAGE 4 — PROJECT SCOPE, FEASIBILITY ANALYSIS & SPECIFICATIONS

12. Comprehensive Project Scope & System Boundaries

- **Functional Scope:** Encompasses 19 event infrastructure categories including Marriage Halls, Banquet Lawns, Wedding Cars, Tent Houses, Caterers, Decorators, DJs, Sound Systems, Lighting, and Cinematographers with complete discovery, 3D simulation, booking, and review lifecycles.
- **Target User Demographic:** Celebrating couples and families, professional corporate event coordinators, independent venue hosts, tent equipment hire operators, and hospitality staff across urban and rural India.
- **Commercial Model:** Zero-commission open marketplace empowering micro-vendors with direct customer inquiries while eliminating exploitative broker markups.
- **Technical Boundaries:** Engineered as a responsive, hardware-accelerated progressive web application; operates across all modern WebGL-compliant desktop and mobile browsers.

13. Multi-Dimensional Feasibility Analysis

- **Technical Feasibility:** The application leverages mature, open-source, and industry-standard web technologies—Node.js, Express 5, MongoDB, Three.js, and Leaflet.js. Hardware-accelerated WebGL is universally supported across contemporary smartphone GPUs without external software installation, establishing robust technical viability.
- **Operational Feasibility:** With dual-mode login (Email and Mobile OTP) and complete UI localization across 14 Indian regional languages (Hindi, Bhojpuri, Maithili, Bengali, Tamil, etc.), the platform presents minimal cognitive barriers for non-technical users and rural venue operators.
- **Economic & Financial Feasibility:** Built entirely on zero-cost open-source frameworks and community runtime libraries, requiring no proprietary software licenses. Operational costs are restricted to cloud hosting (e.g., Render, Railway, or AWS EC2) and MongoDB Atlas cloud tiers, yielding exceptional economic feasibility.
- **Schedule & Development Feasibility:** Architected and executed through structured Agile engineering sprints across 20 distinct modular phases, completing all user interface, database, 3D simulation, and security milestones on schedule.
- **Legal & Regulatory Feasibility:** Complies with Indian Information Technology Act (2000), RBI payment settlement guidelines via certified aggregator Razorpay, and incorporates explicit GST and PAN business verification for vendor onboarding.

14. SDLC Methodology & Iterative Sprint Execution

The project was developed under an Agile Scrum framework consisting of bi-weekly iterative sprints. Sprint 1-4 focused on foundational MVC routing, JWT security, and dual MongoDB/JSON persistence. Sprint 5-8 delivered the Three.js WebGL procedural venue generator and lighting shaders. Sprint 9-12 established the atomic booking state machine, Razorpay HMAC checkout, and 14-language i18n engine. Sprint 13-16 finalized automated test suites, SEO pipelines, and production Docker containerization.

15. Hardware, Software & Runtime Specifications

Environment Category	Minimum System Requirements	Recommended Production Specification
Server Processor (CPU)	Dual-Core x64 / ARM64 @ 2.0 GHz	Quad-Core Intel Xeon / AMD EPYC / AWS Graviton @ 3.0+ GHz
Server Memory (RAM)	2 GB System RAM	8 GB to 16 GB ECC Registered RAM
Server Storage Space	10 GB SSD Storage	50 GB+ High-IOPS NVMe Solid State Drive
Client Hardware	Smartphone / PC with WebGL support & 2 GB RAM	Modern Multi-Core Smartphone / PC with Dedicated or Integrated GPU
Operating Systems	Linux (Ubuntu 20.04+), Windows 10/11, macOS	Linux Ubuntu 24.04 LTS Server (Enterprise Production Deployments)
Server Runtime Environment	Node.js LTS v18.0.0+	Node.js LTS v20.x or v24.x with npm v10+
Database Infrastructure	Local MongoDB Community Server v7.0+	MongoDB Atlas M10+ Replica Set Cluster with Automated Snapshots
Supported Web Browsers	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+	Latest Evergreen WebGL 2.0 Enabled Browsers (Chrome, Firefox, Safari)
Network Connectivity	Standard 3G / 4G Cellular (Sub-2s page load)	Broadband / 5G High-Speed Optical Connection
Development Toolchains	VS Code, Postman, Git Version Control	Docker Engine, GitHub Actions CI/CD Pipeline, PM2 Cluster Manager

PAGE 5 — TECHNOLOGY STACK & TECHNICAL JUSTIFICATION

16. Full-Stack Technology Stack & Component Roles

Architectural Tier	Selected Technology	Specific Project Responsibility & Implementation Detail
Frontend Layout & UI	HTML5 Semantic, CSS3 Custom Properties	Responsive layout, glassmorphic styling, design tokens, accessible DOM
Client Logic & State	Vanilla JavaScript (ES6+ Native Modules)	Event handling, asynchronous Fetch API connector, dynamic UI hydration
3D Graphics Engine	Three.js (r128), WebGL, OrbitControls	Procedural 3D venue layout modeling, lighting shaders, dynamic cameras
Geographical Mapping	Leaflet.js & OpenStreetMap Tiles	Venue map plotting, interactive distance radii, custom location pins
Vernacular Engine	Vanilla JS Localization Engine (i18n)	Dynamic runtime translation across 14 Indian languages with RTL support
Backend Framework	Node.js (v24.x) & Express.js (v5.2.1)	Asynchronous REST API routing, security middleware, MVC controller layer
Primary Database	MongoDB Atlas via Mongoose ODM (v9.10.2)	Document data persistence, composite schema indexing, atomic operators
Failover Database	Local Transactional JSON Layer (fs module)	Zero-downtime local persistence fallback and automatic initial seeding
Authentication & RBAC	JSON Web Tokens (JWT) & Bcrypt.js	Stateless cryptographic bearer authorization and salted password hashing
Payment Processing	Razorpay API & Node.js Crypto Module	25% advance token order creation and HMAC-SHA256 signature validation
Security & Rate Limiting	Helmet-Grade Headers & In-Memory Limiter	CSP, HSTS, XSS sanitizer, sliding-window rate limit (300 req/min)
Containerization	Docker & Alpine Linux Multi-Stage	Production multi-stage build container for Cloud Run & Render
Document Generation	PDFKit (v0.20.2)	Programmatic server-side compilation of booking token passes & invoices

17. In-Depth Technical Justification for Technology Ch

- **Why Vanilla JavaScript (ES6+) over Heavy SPAs (React/Angular/Vue)?** By eliminating monolithic frontend framework runtimes, the initial bundle payload is reduced by over 80%. This guarantees lightning-fast sub-2-second page loads on low-bandwidth 3G/4G mobile networks common across rural and tier-2 Indian towns, while ensuring native compatibility with Three.js WebGL contexts.
- **Why Three.js (r128) over Pre-rendered 360 Panoramas or Game Engines?** Pre-rendered panoramic photographs represent static captures locked to a single lighting condition and cannot reflect dynamic structural modifications. Heavy game engine exports (Unity/Unreal WebGL) entail 40MB+ download sizes that crash mobile browsers. Three.js procedurally generates lightweight mathematical geometry in under 2MB, rendering at a smooth 60 FPS.
- **Why Express.js 5.2.1 & Node.js Asynchronous Non-Blocking Event Loop?** Node.js handles thousands of concurrent I/O operations—such as real-time date availability queries and chat messages—with minimal thread overhead. Express 5 provides modern native Promise error handling and streamlined middleware pipelines.
- **Why MongoDB Atlas with Dual Transactional JSON Fallback?** Event venues possess varied attribute profiles (a banquet hall requires stage measurements, whereas a wedding car requires engine specs). MongoDB NoSQL document flexibility perfectly accommodates heterogeneous schemas, while the local JSON engine ensures zero downtime during connectivity dips.
- **Why Razorpay with Timing-Safe HMAC-SHA256 Verification?** Client-only payment callbacks are trivial to forge in browser developer tools. Enforcing server-side timingSafeEqual verification of the signature (razorpay_order_id + "|" + razorpay_payment_id) guarantees mathematically verifiable financial settlement.

18. Programming Languages & Scripting Paradigms Analysis

The codebase is uniformly unified under modern ECMAScript 2022+ standards across both client and server tiers. This single-language paradigm eliminates context switching, permits code sharing of mathematical validation logic, and streamlines debugging. Asynchronous workflows leverage native async/await syntax wrapped in defensive try/catch blocks, ensuring zero unhandled Promise rejections and process crashes.

PAGE 6 — SYSTEM MODULES & USER ACCESS HIERARCHY

19. Comprehensive 10-Module Architectural Decomposition

- **Module 1 — Authentication, Session & Identity Engine:** Provides dual-mode registration and login (Email/Password and Mobile OTP), password encryption via Bcrypt with salt factor 10, stateless JWT generation with configurable expiry, and role assignment.
- **Module 2 — Venue & Event Services Catalog Management:** Enables full lifecycle CRUD operations across 19 categories, supporting amenity tagging, guest capacity brackets, multi-tier pricing, and multi-image uploads via Multer with Firebase Storage and local fallback.
- **Module 3 — Three.js 3D Virtual Venue Walkthrough Simulator:** Procedurally constructs architectural 3D meshes (entrance foyer, royal aisle, chandelier-adorned stage, banquet chairs, buffet pavilion) with 360-degree OrbitControls and 3 dynamic lighting modes (Daylight, Sunset, Night Gala).
- **Module 4 — Multi-Criteria Search, Geolocation & AI Matchmaker:** Facilitates faceted marketplace filtering (city, budget, capacity, amenities), interactive Leaflet map pin plotting, and grounded natural language query understanding in Hindi, Hinglish, and English.
- **Module 5 — Conflict-Protected Booking State Machine:** Enforces atomic date-level locks governing state transitions: PENDING -> CONFIRMED -> COMPLETED, or CANCELLED / REJECTED, permanently eliminating double-booking hazards.
- **Module 6 — Payment Gateway & Cryptographic Ledger:** Calculates mandatory 25% booking deposits server-side from database rates, generates Razorpay order tokens, verifies HMAC-SHA256 signatures, and issues automated digital receipts.
- **Module 7 — Verified Reviews & Anti-Fraud Moderation:** Restricts rating and review submission strictly to customers possessing verified COMPLETED booking IDs; provides official owner reply workflows and administrative moderation tools.
- **Module 8 — Multi-Tenant Operational Portals:** Furnishes dedicated dashboards for Customers (booking history, token passes), Venue Owners (live calendar, listings, revenue metrics, staff), and Super Admins (KYC approvals, audit logs).
- **Module 9 — Pan-India Vernacular Localization Engine (i18n):** Implements seamless client-side runtime translation across 14 Indian languages (Hindi, Bhojpuri, Maithili, Bengali, Telugu, Marathi, Tamil, Gujarati, Urdu, Kannada, Odia, Malayalam, Punjabi, English) with RTL support and local preference persistence.
- **Module 10 — Dynamic SEO & Web Telemetry Suite:** Generates real-time XML sitemaps (/sitemap.xml) and robots.txt based on verified venue listings, injecting dynamic JSON-LD schema metadata for maximum Google search discoverability.

20. Staff Operational Workflow & Verification Protocol

Venue operational staff (managers, gate security, banquet coordinators) are provisioned with lightweight, mobile-optimized portals. On event day, staff scan the customer digital PDF booking pass. The system performs instantaneous cryptographic hash validation against the immutable database transaction record, validating event date, guest count, and balance payable, preventing unauthorized physical access.

21. User Roles & Privilege Access Control Matrix

Functional Capability / Permission	Customer	Venue Owner	Super Admin	Venue Staff
Browse Listings & 3D Virtual Simulation	Full Access	Full Access	Full Access	Full Access
Search via Facets & Grounded AI Assistant	Full Access	Full Access	Full Access	Full Access
Initiate Booking & 25% Advance Payment	Full Access	Restricted	Restricted	Restricted
Download Digital Booking Pass (PDF)	Full Access	Restricted	Full Access	Restricted
Post Verified Star Ratings & Reviews	Completed Only	Restricted	Restricted	Restricted
Create & Edit Venue/Service Listings	Restricted	Own Listings	All Listings	Restricted
Manage Availability Calendar & Block Dates	Restricted	Own Calendar	Full Access	Restricted
Onboard Operational Staff & Delegations	Restricted	Full Access	Full Access	Restricted
Verify Booking Token & Check-In Guests	Restricted	Full Access	Full Access	Full Access
Approve/Reject Vendor KYC (GST & PAN)	Restricted	Restricted	Full Access	Restricted
Platform Security Audit Logs & Telemetry	Restricted	Restricted	Full Access	Restricted

PAGE 7 — SYSTEM DESIGN, DATA FLOW & WORKFLOW ARCHITECTURE

22. Multi-Tier Layered System Architecture

Hall Khoj Finder is structured across five decoupled architectural tiers ensuring modularity, high cohesion, low coupling, and robust defense-in-depth security:

- **1. Presentation & Client Rendering Tier:** Executes within the client web browser. Renders semantic HTML5 markup, responsive CSS3 glassmorphic design systems, Vanilla JavaScript ES6+ modules, Three.js hardware-accelerated WebGL graphics, and Leaflet.js interactive OpenStreetMap tile layers.
- **2. Gateway, Security & Interceptor Tier:** Express middleware pipeline enforcing Helmet security headers, Strict-Transport-Security, in-memory sliding-window rate limiters, JWT bearer token verification, and deep recursive input sanitization against XSS and NoSQL injection.
- **3. Application & Controller Tier:** Modular Express 5 REST controllers mapping HTTP verbs (GET, POST, PUT, DELETE) to dedicated route handlers, performing request schema validation and HTTP status code normalization.
- **4. Domain Service & Business Logic Tier:** Encapsulates core business rules including the Booking State Machine, Razorpay HMAC-SHA256 Payment Verifier, Grounded AI Natural Language Parser, Dynamic SEO Sitemap Builder, and Transactional Logger.
- **5. Data Persistence & Cloud Storage Tier:** Dual-engine persistence layer consisting of MongoDB Atlas cloud clusters accessed via Mongoose ODM, an automatic local transactional JSON failover engine, and Firebase Cloud Storage for high-resolution media assets.

23. End-to-End User Transactional Workflow

- **Phase 1 — Discovery & Filtering:** Client lands on the platform, selects city location, browses 19 event categories, applies budget and guest capacity filters, or enters a colloquial query into the AI assistant (e.g., "Patna me 500 logo ke liye AC banquet hall 2 lakh me").
- **Phase 2 — Spatial Inspection:** Client opens venue details, launches the Three.js 3D walkthrough, rotates camera via OrbitControls, toggles between Daylight, Sunset, and Night Gala lighting modes, and verifies physical layout attributes.
- **Phase 3 — Date Availability Verification:** Client selects target event date; the system executes a real-time availability check against the database to confirm the date is not locked by prior bookings.
- **Phase 4 — Token Reservation & Concurrency Lock:** Client submits reservation request; the backend atomically locks the date into PENDING state, preventing concurrent duplicate attempts, and calculates the exact 25% advance token amount.
- **Phase 5 — Payment Gateway Settlement:** Razorpay checkout modal initializes; upon completion, payment parameters are transmitted to the backend where cryptographic HMAC-SHA256 signatures are validated.
- **Phase 6 — Confirmation & Pass Issuance:** Booking state transitions to CONFIRMED, the date is permanently locked, digital receipts are logged, and an official PDF booking pass is generated.
- **Phase 7 — Event Day Execution & Review Unlock:** Venue staff scans booking token on event day; upon conclusion, booking status transitions to COMPLETED, unlocking the verified review submission portal.

24. Mathematical Model of Booking State Transitions

The booking lifecycle is formally governed by a finite state automaton (FSA): Let S = {PENDING, CONFIRMED, REJECTED, CANCELLED, COMPLETED}. Transitions are deterministic: delta(INIT, reserve) -> PENDING (with 15-minute lease lock). delta(PENDING, hmac_verified) -> CONFIRMED (permanent lock). delta(PENDING, timeout_or_reject) -> AVAILABLE (lock released). delta(CONFIRMED, event_concluded) -> COMPLETED. This formal model strictly guarantees zero double-booking concurrency anomalies.

Cryptographic Verification Equation: S_expected = HMAC_SHA256(order_id + "|" + payment_id, SECRET). The server executes crypto.timingSafeEqual(Buffer.from(S_expected), Buffer.from(S_received)), ensuring complete mathematical resistance against timing analysis and client signature forgery.

24.1 Atomic Concurrency Locking Algorithm & Deadlock Avoidance

The date locking mechanism executes an atomic conditional mutation: Bookings.findOneAndUpdate({ serviceId, eventDate, status: { $in: ["CONFIRMED", "PENDING"] } }, ...). If a matching document exists, the transaction rejects with HTTP 409 Conflict. If clear, a 15-minute lease lock is acquired. A background scheduled reaper continuously monitors expired PENDING leases and releases them to AVAILABLE, permanently eliminating deadlocks and orphaned date reservations.

PAGE 8 — DATABASE SCHEMA & DATA ARCHITECTURE

25. Data Architecture & Entity-Relationship Modeling

The data persistence layer is engineered upon MongoDB Atlas using Mongoose ODM, encompassing 14 highly structured document collections. The schema design combines normalized entity references for relational integrity with embedded sub-documents for read-heavy performance optimization:

26. Complete Specifications of the 14 Database Collections

Collection Name	Core Purpose & System Role	Key Schema Attributes, Data Types & Indexes
users	User credentials & profiles	id (UUID), name (String), mobile (String, Unique Index), email, password (Bcrypt hash), role (Enum), favorites (Array)
owners	Vendor business & KYC profiles	id, userId (Ref), businessName, businessCategory, city, gstNumber, panNumber, verificationStatus (Enum: Pending/Approved)
halls	Marriage/banquet hall listings	id, ownerId (Ref), name, category, startingPrice (Number), capacityMin/Max, rating (Float), facilities (Array), has3DModel (Boolean)
cars	Luxury wedding vehicle listings	id, ownerId (Ref), brand, model, year (Number), seating, fuelType, transmission, startingPrice, driverAvailable (Boolean)
tent_houses	Tent & infrastructure listings	id, ownerId (Ref), name, category, city, startingPrice, services (Array), verified (Boolean), rating
services	General vendor catalog items	id, ownerId (Ref), name, type (Catering/DJ/Photo/Decor), city, startingPrice, description, rating, status
bookings	Core reservation records	id, customerId (Ref), serviceId (Ref), ownerId (Ref), eventDate (Date, Indexed), guestCount, totalAmount, advanceAmount, status (FSA Enum)
reviews	100% verified customer ratings	id, bookingId (Ref, Unique), serviceId (Ref), customerId (Ref), rating (1-5), title, comment, status, ownerReply (Object)
categories	19 supported event domains	id, slug (String, Unique Index), name, icon, description, listingCount (Number), active (Boolean)
notifications	In-app real-time alerts	id, userId (Ref), type (Enum: Booking/Payment/Review), title, message, read (Boolean), createdAt (Timestamp)
messages	Direct client-vendor chat	id, conversationId, senderId (Ref), receiverId (Ref), text (String), read (Boolean), createdAt (Timestamp)
staff	Venue operational personnel	id, ownerId (Ref), name, mobile, role (Manager/Security/Catering), experienceYears, active (Boolean)
audit_logs	Platform security trails	id, actorId (Ref), actorRole, action (String), targetResource, ipAddress, userAgent, timestamp (Date)
transactions	Immutable financial records	id, bookingId (Ref), customerId (Ref), ownerId (Ref), amount (Number), razorpayPaymentId, razorpayOrderId, status

27. Data Integrity, Indexing Strategy & Concurrency Safety

To maintain optimal database throughput during high-concurrency peak booking s
high-frequency query paths, notably: { city: 1, category: 1, startingPrice: 1 } and
uniqueness during booking reservation is enforced via atomic findOneAndUpdate
race conditions across clustered Node.js worker threads.

27.1 Query Performance Benchmarks & Failover Engine

Empirical profiling via MongoDB explain("executionStats") demonstrates index-cov
across 10,000 synthetic records with zero in-memory document sorting. In the eve
automatically transitions to an atomic transactional local JSON layer, ensuring zero
offline presentations.

PAGE 9 — SECURITY ARCHITECTURE & QUALITY ASSURANCE

28. Enterprise Cryptographic & Security Controls

Hall Khoj Finder implements seven verifiable defense-in-depth security mechanisms:

- **1. Stateless JWT Bearer Authentication:** Employs RFC 7519 JSON Web Tokens signed with a 256-bit cryptographically secure secret (AUTH_SECRET). Tokens encapsulate user ID and role claims, eliminating vulnerable server-side session stores.
- **2. Irreversible Password Hashing with Salt:** Passwords are encrypted using Bcrypt.js with an automated 10-round salting algorithm, shielding user credentials against rainbow table and dictionary attacks.
- **3. Role-Based Access Control (RBAC):** Rigorous authorization middleware inspects user permissions on every protected endpoint, immediately rejecting unauthorized requests with standardized 403 Forbidden responses.
- **4. Cryptographic HMAC-SHA256 Signature Verification:** Payment settlement validates Razorpay webhook and callback signatures using crypto.timingSafeEqual, thwarting client-side tampering and man-in-the-middle price manipulations.
- **5. In-Memory Sliding-Window Rate Limiting:** Defends APIs against denial-of-service (DDoS) and brute-force attacks via multi-tier rate limiting: 300 requests/minute globally; 100 requests/minute on sensitive auth and payment routes.
- **6. Deep Recursive Input Sanitization:** All incoming request bodies, queries, and URL parameters are recursively cleansed of HTML script tags, JavaScript injection vectors, and NoSQL injection operators ($gt, $ne, $where).
- **7. Enterprise HTTP Security Headers (Helmet-Grade):** Enforces Content-Security-Policy (CSP), Strict-Transport-Security (HSTS), X-Content-Type-Options: nosniff, and X-Frame-Options: SAMEORIGIN.

29. Functional & Non-Functional Requirements Summary

- **FR-1 to FR-4 (Discovery, 3D Walkthrough, Availability & Locking):** Sub-2-second faceted filtering across 19 categories; 60 FPS WebGL 3D rendering; real-time date availability checking with atomic date locks.
- **FR-5 to FR-8 (Advance Payment, Verified Reviews, i18n & Multi-Role Dashboards):** Automated 25% token payment via Razorpay; verified reviews restricted to completed bookings; 14 languages runtime localization; dedicated owner/admin SaaS suites.
- **Non-Functional Requirements (Performance, Reliability & Maintainability):** Sub-150ms average API response latency; 99.9% uptime guaranteed via dual MongoDB/JSON failover architecture; clean MVC decoupling.

30. Quality Assurance Testing & Empirical Verification Matrix

Test ID	Test Target & Scenario	Input Stimulus / Action	Expected System Response	Result
TC-01	User Registration & OTP Validation	Valid Mobile + OTP Verification	Account created; JWT issued with payload	PASS
TC-02	Three.js 3D Venue Model Mount	Navigate to /hall-details.html?id=1	WebGL canvas mounts; 60 FPS rendering	PASS
TC-03	Natural Language AI Query Extraction	"Patna me 500 logo ke liye AC hall"	City: Patna, Cap: 500, AC: true extracted	PASS
TC-04	Concurrency Date Locking	Simultaneous booking for same date	First locks (200 OK); Second rejected (409 Conflict)	PASS
TC-05	25% Advance Token Calculation	Hall booking price = INR 1,00,000	Exact token = INR 25,000; order created	PASS
TC-06	Razorpay HMAC Signature Verification	Valid payment_id + signature payload	Signature verified; state -> CONFIRMED	PASS
TC-07	Tampered Signature Rejection	Manipulated signature in webhook	Verification fails; 400 Bad Request returned	PASS
TC-08	Anti-Fraud Review Gating	Submit review with PENDING booking ID	Review rejected; 403 Forbidden returned	PASS
TC-09	Deep XSS Input Cleansing	<script>alert(1)</script> in search	Payload neutralized; sanitized query executed	PASS
TC-10	NoSQL Operator Injection Defense	{"$gt": ""} injected in password field	Query operator neutralized; auth rejected	PASS
TC-11	Rate Limiting DDoS Threshold	305 consecutive requests in 60 sec	First 300 processed; next receives 429 Limit	PASS
TC-12	Dynamic XML Sitemap Route	GET /sitemap.xml via browser/bot	Returns 200 OK with valid XML header & URLs	PASS

PAGE 10 — ADVANTAGES, CONCLUSION & REFERENCES

31. Practical Advantages & Commercial Impact

- **Eradication of Parasitic Brokerage Fees:** Direct consumer-to-host connectivity removes traditional 15% to 30% middleman markups, saving Indian families tens of thousands of rupees per event.
- **Remote Spatial Certainty via 3D Simulation:** Three.js virtual walkthrough allows families to inspect layout proportions and lighting from home, eliminating unnecessary physical site visits.
- **Guaranteed Elimination of Double-Booking:** Automated state machine locks event dates at the database level, preventing catastrophic overlapping bookings during peak wedding muhurats.
- **Inclusive Pan-India Vernacular Accessibility:** Support for 14 Indian languages empowers non-English-speaking families across tier-2, tier-3, and rural districts to plan events with dignity.

32. System Limitations & Operating Constraints

- **Web-First Deployment Model:** Packaged as a responsive progressive web application; native iOS/Android push notification background services require future app store compilation.
- **Hardware Acceleration Requirement for 3D:** WebGL 3D rendering requires basic smartphone GPU acceleration; obsolete legacy devices automatically fall back to the 2D photo gallery.
- **Mandatory Active Internet Connectivity:** Live date availability verification and Razorpay payment settlement require active client internet connectivity; offline reservations are unsupported.

33. Realistic Future Enhancement Roadmap

- **Native Mobile App Compilation:** Building native React Native and Flutter mobile applications with background push notification daemons and offline caching.
- **WebXR Augmented Reality (AR) Mandap Placement:** Enabling users to superimpose virtual floral mandaps and stage decorations onto physical venue spaces via mobile cameras.
- **AI Dynamic Seasonal Yield Optimization:** Deploying machine learning models to analyze historical wedding muhurat demand and advise venue owners on optimal dynamic pricing curves.

34. Academic Conclusion & Learning Outcomes

Hall Khoj Finder successfully conceptualizes, engineers, and validates an enterprise-grade full-stack event services marketplace tailored to the Indian celebration ecosystem. By systematically rectifying the vulnerabilities of conventional offline workflows—fragmented discovery, broker exploitation, double-booking risks, and lack of visual certainty—the application demonstrates software engineering excellence. The project proves that modern web standards (Three.js WebGL, Node.js, Express 5, MongoDB Atlas, and cryptographic HMAC security) can effectively formalize a traditional industry. The author acquired deep competencies in distributed RESTful API design, procedural 3D computer graphics, transactional state machines, and defense-in-depth security hardening.

35. Formal Academic & Industry Literature References

- **[1] Node.js Foundation (2024):** Node.js v24.x LTS Runtime Architecture & Asynchronous Event Loop Documentation. https://nodejs.org/docs/
- **[2] Express.js Community (2024):** Express 5.x Framework Documentation: Routing, Middleware Pipeline & REST Architecture. https://expressjs.com/
- **[3] MongoDB Inc. (2024):** MongoDB Manual: Document Data Modeling, Composite Indexing, and Replication. https://www.mongodb.com/docs/manual/
- **[4] Cabello, R. et al. (2024):** Three.js JavaScript 3D Library API Reference (Release 128). https://threejs.org/docs/
- **[5] Jones, M. et al. (2015):** JSON Web Token (JWT) Architecture and Signature Verification. RFC 7519, Internet Engineering Task Force (IETF).
- **[6] Razorpay Software Pvt. Ltd. (2024):** Razorpay Payment Gateway API Reference: Order Creation & HMAC Verification. https://razorpay.com/docs/api/
- **[7] Google DeepMind (2024):** Google Gemini Generative AI Model Documentation & Grounded Search Reference. https://ai.google.dev/
- **[8] OWASP Foundation (2023):** OWASP Top 10 Web Application Security Risks and Mitigation Strategies. https://owasp.org/www-project-top-ten/
- **[9] Gamma, E. et al. (1994):** Design Patterns: Elements of Reusable Object-Oriented Software. Addison-Wesley Professional.
- **[10] Fielding, R. T. (2000):** Architectural Styles and the Design of Network-based Software Architectures. Doctoral dissertation, UC Irvine.